Title : A Multicore and Edge TPU-Accelerated Multimodal TinyML System for Livestock Behavior Recognition

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A MULTICORE AND EDGE TPU-ACCELERATED MULTIMODAL TINYML SYSTEM FOR LIVESTOCK BEHAVIOR RECOGNITION

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

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ABSTRACT

The rapid advancements of Tiny Machine Learning (TinyML) and Edge Computing have made possible the creation of low-cost and efficient intelligent monitoring systems. In this paper, we describe a multicore and Edge TPU accelerated multimodal TinyML system that recognizes livestock behaviors.

The proposed system monitors and classifies livestock activities in real time, utilizing embedded microcontrollers, sensors, and machine learning models. The system utilizes both accelerometer sensor data and camera image inputs to detect animal behavior, including resting, moving, grazing, and abnormal activities.

TinyML models were optimized and deployed on microcontrollers with Edge TPU acceleration to enable fast and energy-efficient inference. The proposed system utilized multimodal fusion methods to improve both accuracy and reliability when compared to traditional single-modality systems.

Wireless communication was also employed to transmit activity information and alerts from the system to allow remote monitoring of livestock by farmers, without reliance upon cloud-based infrastructure. This method reduces latency, increases user privacy, and supports reliable operation in areas of limited or no internet access.

The proposed system represents an efficient, scalable, and cost-effective means of implementing smart livestock monitoring systems that enhance farm productivity, decrease manual labor, and provide early detection of abnormal animal behavior supporting modern intelligent agriculture.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	i
	ACKNOWLEDGEMENT	ii
	TABLE OF CONTENTS	iii
CHAPTER 1	INTRODUCTION	
1.1	Introduction	
1.2	Problem Statement	
1.3	Objectives of the Project	
1.4	Scope of the Project	
1.5	Organization of the Report	
CHAPTER 2	LITERATURE SURVEY	
2.1	Introduction	
2.2	Review of Existing Systems	
2.3	Limitations of Existing Systems	
2.4	Proposed System Overview	
CHAPTER 3	SYSTEM ANALYSIS	
3.1	Existing System	

3.2	Proposed System	
3.3	System Requirements	
CHAPTER 4	SYSTEM DESIGN	
4.1	System Architecture	
4.2	Data Flow Diagram	
4.3	UML Diagrams	
CHAPTER 5	IMPLEMENTATION	
5.1	Tools and Technologies Used	
5.2	Module Description	
5.4	Screenshots	
CHAPTER 6	RESULTS AND DISCUSSION	
6.1	System Testing	
6.2	Output Results	
CHAPTER 7	CONCLUSION AND FUTURE WORK	
7.1	Conclusion	
7.2	Future Enhancements	
	REFERENCES	
	APPENDIX A – BASE PAPER	

BASE PAPER DETAILS

Title:
A Multicore and Edge TPU-Accelerated Multimodal TinyML System for Livestock Behavior Recognition

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1.1 Introduction

Agriculture has significant contributions to the global economy, and is a leading area in the livestock industry as it relates to food production and rural development. Properly monitoring the behaviors of livestock is critical to the health of the animals and the efficiency of production. However, historically, farmers monitored livestock behavior by direct observation, which was time consuming, required a lot of labor, and had a high potential for human error.

Due to advancements in IoT, Embedded Systems, and Machine Learning; it now is feasible to create automated livestock monitoring systems. Tiny Machine Learning (TinyML) is an emerging technology that allows for machine learning models to be executed on low-power microcontrollers and other embedded devices. TinyML enables real-time data analysis using the local device's processing capabilities and does not require cloud computing, thereby decreasing latency, power usage, and dependence on internet connectivity.

This project includes the development of a multicore and Edge TPU accelerated multimodal TinyML-based system for recognizing livestock behavioral patterns. This system collects data from animals using sensors such as cameras and accelerometers and uses optimized machine learning models to identify the behavior of the animals based on the collected data. In addition to collecting data from a single source, the system uses a multimodal approach by combining data from multiple sources to improve the reliability and accuracy of the results.

The system is designed to operate on microcontrollers that utilize Edge TPU acceleration to enable rapid and efficient real-time inference. Data collected by the system is transmitted wirelessly to the farmer, providing them with the ability to remotely monitor the animals' behavior and receive alerts and activity information regarding the animals. This provides an efficient, cost effective, and scalable solution for intelligent livestock monitoring and smart agriculture.

1.2 Problem Statement

Livestock farming is a field of continuous animal behavior evaluation for the health, security and productivity of the animals. Traditional livestock behavior evaluation tools are primarily based on farmer's direct observations. Farmer's observations require a lot of time and labor and do not provide a reliable assessment of the behavior of the animals. Farmers have difficulty evaluating the behavior of all the animals when they have a very large number of animals in one location, and cannot evaluate them continuously.

Current livestock behavior monitoring systems are typically dependent upon cloud computing and powerful hardware. The dependence upon cloud computing results in higher costs, increased latency and a reliance upon consistent and reliable Internet access. Internet access in rural locations may be unreliable or non-existent; therefore cloud-based livestock behavior monitoring systems are not ideal for this environment. Many current livestock behavior monitoring systems also utilize only a single type of data source (i.e., sensor(s) or camera(s)) to monitor the animals. This reduces the accuracy and reliability of the behavior recognition capabilities of the system.

Therefore, there is an urgent need for a livestock behavior monitoring system that is low-cost, efficient and provides real-time information about the behavior of the animals. The new system should be able to recognize animal behavior utilizing multiple types of data sources (e.g., video feed from cameras and/or sensor readings) and provide timely notifications to the farmer regarding changes in animal behavior.

The challenges outlined above were addressed through the development of a multicore and Edge TPU-accelerated multimodal TinyML system that can run directly on embedded microcontrollers. This allows for the development of a livestock behavior monitoring system that provides real-time evaluations of animal behavior with improved efficiency, accuracy, and reliability.

1.3 Objectives of the Project

This project aims to provide an advanced livestock behavior recognition system that utilizes Tiny Machine Learning (Tiny ML) with Edge TPU technology for an optimized smart livestock monitoring system that uses real time embedded microcontroller based activity tracking of livestock.

This project has three main objectives:

• Develop a TinyML-based System for Real-Time Livestock Behavior Recognition.

• Collect Data from Multiple Sources (e.g., Accelerometer Sensors & Camera Images).

• Implement Optimized Machine Learning Models on Resource-Constrained Microcontrollers.

• Use Edge TPU to Improve Inference Speed and System Performance.

• Implement Multimodal Data Fusion Techniques to Enhance Accuracy and Reliability of Behavior Recognition.

• Develop Wireless Communication Mechanism to Send Alerts and Activity Information.

• Reduce Cloud Computing Dependency by Performing On-Device Inference.

• Provide Low-Cost, Scalable, Efficient Solution for Smart Livestock Monitoring.

1.4 Scope of the Project

The scope of the project's focus will be developing a Smart Livestock Behavior Detection System using Tiny Machine Learning (TinyML) and Edge TPU Acceleration. Embedded microcontrollers, sensors and Machine Learning Algorithms will be used to detect animal behavior including Rest, Movement, Grazing and Abnormal behavior.

Sensor and Video Data will be collected from cameras and accelerometers that are mounted on animals. The data collected will be processed by TinyML optimized models that run on microcontrollers in Real-Time.

Multi-modal Data Fusion will be utilized to improve performance by processing and merging data from multiple different data collection methods. Since the system operates independently of Cloud Computing and Internet Access it can operate effectively in rural areas and remote areas with little to no Internet Access. Alert Messages and Activity Information will be sent to Farmers via wireless means for effective Monitoring.

The primary objective of this undertaking was to develop and evaluate an affordable, efficient, scalable embedded system that can effectively monitor livestock.

Additionally, the system has the potential for growth in the future through the addition of new hardware (sensors), software (machine learning models) and functionality (monitoring).

1.5 Organization of the Project Report

This project report is organized into three distinct chapters to clearly explain how TinyML with Edge TPU acceleration can be used to develop a livestock behavior recognition system.

Chapter 1 contains the Introduction that includes the background of the project, problem statement, objectives, scope and the organization of the report.

Chapter 2 represents the Literature Survey that includes an overview of the available literature for livestock monitoring systems, TinyML applications, and machine learning techniques applied for animal behavior recognition.

Chapter 3 describes the System Analysis, and it is divided into four sections: the current system, the proposed system, the system requirements, and the feasibility study.

Chapter 4 represents the System Design, and includes three main areas: system architecture, data flow diagrams, UML diagrams, and the general design of the system.

Chapter 5 contains the description of the Implementation of the system, and covers the tools and technology used, each module description, algorithms, and the process of developing the system.

Chapter 6 includes the Results and Discussion, and includes the testing of the system, the results obtained from the output of the system, and the performance evaluation.

Chapter 7 represents the Conclusion and Future Work, and summarizes the results of the project and proposes potential improvements for the future.

Finally, the report has two supplementary parts: References and Appendix, and includes the base paper and other relevant supporting documentation.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Livestock Monitoring has gained a lot of interest in the recent years due to both the positive impact on livestock health as well as agricultural productivity and efficiency.
Traditional livestock monitoring methods are based upon manual observations by farmers, which are very time consuming and labor intensive. Due to the rapid development of embedded systems, IoT technologies and Machine Learning, new methods of automating livestock monitoring have emerged that allow for greater levels of efficiency and precision.

Tiny Machine Learning (Tiny ML) is an emerging field of research that enables machine learning algorithms to operate at extremely low power levels and can be executed on small micro controllers and/or embedded system platforms.
This will enable the operation of real time data processing directly on device, eliminating reliance on Cloud Computing and therefore reducing latency, power usage and dependency on internet connectivity.
Therefore, this new field of research may be applied to Smart Agriculture applications.

There are a number of different research studies that propose livestock monitoring systems that utilize various forms of sensors, cameras and machine learning based models. Each system utilizes one or more of the

following components: accelerometers; cameras; and/or wireless communication to obtain and analyze data on animal behaviors. Additionally, each system utilizes one or more of the following machine learning based techniques: Convolutional Neural Networks (CNN); Long Short-Term Memory (LSTM) models; and/or object detection based models to identify individual animal behaviors.

The purpose of this chapter is to review existing livestock monitoring systems, TinyML-based application, and machine learning techniques utilized in the recognition of behavior in livestock. The limitations of current systems will be discussed, along with the necessity of developing an efficient, cost-effective, and real-time embedded livestock monitoring solution.

2.2 Review of Existing Systems

To date several livestock monitoring systems have been created through various IoT, sensor and machine learning tools to aid in both improving farm management and monitoring of animal health. The use of wearable sensors, cameras and other types of wireless communication devices enable the collection of data regarding animal behavior. Sensor-based monitoring systems utilize accelerometers, temperature sensors and GPS modules to monitor animal movement and activity. These types of monitoring systems are capable of identifying animal behaviors including resting, walking and grazing; in addition these types of systems utilize machine learning algorithms such as SVM, decision trees and neural networks to categorize animal behavior based on the collected sensor data. Although sensor-based monitoring systems are energy efficient and relatively inexpensive they tend to lack accuracy when utilized by themselves.

Vision-based monitoring systems utilize cameras and computer vision techniques to identify and track animals. CNNs and object detection models

such as YOLO are widely utilized for the identification and tracking of objects in images. Vision-based monitoring systems also provide visual information to enhance the accuracy of monitoring animal behavior; however, these systems are computationally intensive and therefore expensive to implement in a practical sense.

Cloud-based monitoring systems utilize IoT devices to collect data which is then transmitted to cloud-based servers for processing. Cloud-based monitoring systems offer greater processing capabilities and larger storage capacities than traditional non-cloud based systems; however, cloud-based monitoring systems rely heavily upon reliable access to the internet. Internet reliability issues in rural environments can lead to delayed responses and decreased overall system performance.

In recent years, TinyML-based systems have been developed to allow machine learning models to be executed locally on microcontrollers. This provides a means to perform real-time monitoring with less reliance on cloud-based services. A number of current systems continue to rely on singular data sources and do not effectively integrate multimodal data to achieve higher levels of accuracy.

As a result of the above discussion, there exists a demand for an effective livestock monitoring system which utilizes multiple data sources to enable real-time behavior recognition with the aid of embedded systems.

2.3 Limitations of Existing Systems

Although there have been many other livestock monitoring systems created with sensors, cameras, and machine learning, many of these systems have

significant limitations which limit both their functionality and applicability to larger scale farm operations.

One major limitation is that many of the current systems rely on either fully manual monitoring, or at best semi-automated monitoring systems, requiring an operator to intervene throughout the monitoring process. As such, this significantly limits the efficiency of the monitoring process and would be impractical for larger scale farm operations where there is much greater need for automation of all aspects of monitoring and management.

Another significant limitation is that cloud based systems require constant internet connectivity to analyze and process data. Many rural and remote locations do not have reliable access to the internet, thus causing delays in transmitting data and reducing the effectiveness of the system.

In addition to the limitations mentioned above, vision based systems require high levels of computation and therefore require very expensive hardware such as Graphics Processing Units (GPUs) or High Performance Processors (HPPs). Both of these types of hardware increase the total cost of the system, as well as the power required to operate the system, thus limiting the ability to deploy the system on lower cost embedded devices.

Additionally, sensor based systems utilizing only accelerometer data may not accurately identify or classify all possible behaviors of animals. The lack of multiple data sources used to make classifications reduces the overall accuracy of behavior identification.

Finally, most existing systems are not designed with resource constrained microcontrollers in mind, nor are they capable of performing real time processing effectively. These systems typically consume more power than necessary, and require additional processing systems to support operation, thus increasing the overall cost of the system.

2.4 Proposed System Overview

In order to address the shortcomings of current livestock monitoring systems, the goal of this project is to develop an Edge TPU-accelerated and multi-core based multimodal TinyML-based livestock behavior recognition system. The objective of the proposed system is to enable the real time monitoring and classification of animal behavior using microcontrollers as well as optimized machine learning algorithms.

The proposed system will be able to capture data related to movement from the accelerometer (movement) sensor(s) as well as visual data from the camera module(s) of the animal(s). A multimodal sensing approach enhances both the reliability and accuracy of behavior recognition when compared to systems which rely solely on a single data source.

The data captured by the sensors will be processed by TinyML models running directly on microcontrollers. The models have been optimized to operate in an energy-efficient manner and consume very little power on resource constrained platforms such as those found in edge devices.

In addition, the system includes a wireless transmission capability to send notifications and data related to activity to the farmer's mobile device. This allows for remote monitoring to occur without requiring either the cloud or internet access. Due to the fact that the system can function independently, the proposed system is suitable for remote and/or rural locations.
The proposed system provides an effective, scalable, and cost-effective platform for intelligent livestock monitoring; increases farm productivity; decreases manual labor requirements; and enables earlier identification of unusual behavior in animals.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

Existing livestock monitoring technologies have primarily relied upon either manual visual inspection, sensor-based technologies, or cloud-based technology in an effort to measure a variety of aspects about livestock health and activity.

In traditional livestock production, farmers rely upon manual visual inspections of the behavior of their livestock to identify any potential health problems or abnormal behaviors of the animals. While this method allows for the opportunity for farmers to visually inspect the animals at regular intervals, the process is labor intensive and is only feasible for smaller scale operations due to the extensive amount of time that would be required to perform these types of observations.

Sensor-based technologies utilize various sensors including, but not limited to, accelerometers, GPS modules, and temperature sensors which are attached to the animals in order to monitor the movement and health of the animals. Sensor-based technologies collect data regarding the movement and health of the animals and then utilize machine learning algorithms in order to classify the animal's behavior. Sensor-based technologies offer lower costs and energy efficiency compared to many other forms of technologies; however, sensor-based technologies may not produce accurate results if utilized independently, since they typically only gather information based upon animal movement.

Visual-based livestock monitoring systems utilize cameras along with computer vision technologies in order to track and recognize the livestock within the camera's view. Computer vision technologies utilize machine learning models such as Convolutional Neural Networks (CNN), and object

detection technologies in order to interpret the images captured by the camera and determine what type of activity the animals are performing. Visual-based systems typically produce more accurate results than sensor-based systems; however, they also require significantly more computing resources and often require the utilization of very costly hardware.

Cloud-based livestock monitoring systems utilize Internet of Things (IoT) devices in order to collect data from the livestock, and then transmit the collected data to cloud servers where the data is processed and analyzed. Cloud-based systems offer high levels of processing capability; however, the systems' ability to function effectively depends upon reliable access to internet connectivity. In many rural locations, poor quality and unreliable internet connectivity can result in slower response times and reduced overall system performance.

Collectively, current livestock monitoring systems have several limitations that include high operating costs, high power consumption, dependency on cloud-based systems for data processing, and lack of real-time processing capabilities on embedded systems.

3.2 Proposed System

The proposed system is a multimodal (multisensor) TinyML (Tiny Machine Learning) system for real-time livestock behavior recognition. The multimodal TinyML system has been accelerated by an Edge TPU (Google’s AI accelerator), which utilizes multiple cores to increase the processing speed of data collected from multisensor devices in real time.

The system collects data from three types of sensors: accelerometer sensors that are attached to livestock to measure movements; camera modules that capture visual data about the animals and their environments; and wireless transmission of activity data and alerts to farmers via wireless networks.

Each sensor type captures unique data related to the behavior of the livestock. The inclusion of multiple sensor types increases the accuracy of the behavior recognition process as well as the reliability of the data captured.

Because of the low computational resources available on each microcontroller, TinyML models are optimized to run in low-power modes on those microcontrollers. Additionally, Edge TPUs can accelerate model inference to provide faster inference times than the original unaccelerated models.

This capability to perform inference locally, at the edge of the network, eliminates the need for remote access to a cloud service or server to perform analysis. In addition, it increases the speed at which results can be obtained and decisions made regarding abnormal behaviors of the livestock being monitored.

The system also includes wireless communication capabilities that enable activity data and alerts to be transmitted wirelessly to farmers' mobile phones or computers.

This ability to send alerts or data to farmers’ mobile phones or computers enables farmers to monitor livestock remotely and take action when abnormal behaviors are detected.

Overall, the proposed system is a cost-effective, efficient, and scalable solution for intelligent livestock monitoring. Additionally, the proposed system will reduce dependence on cloud-based services and infrastructure for data processing, improve real-time processing capabilities and enhance the overall efficiency of livestock management.

3.3 System Requirements

The requirements of the system for developing and testing a livestock behavior recognition system with TinyML and machine learning are defined in this section as the hardware and software needs of the system.

Hardware Requirements:

• A computer or laptop for developing and testing the system.

Software Requirements:

• Programming language: Python
• Machine Learning framework: TensorFlow Lite (Tiny ML)
• Development environment: Visual Studio Code or Jupyter Notebook
• Operating Systems: Windows, Linux or macOS

Functional Requirements:

• Use of an image dataset of livestock.
• Training of the Tiny ML model.
• Recognition of the livestock's behavior.
• Displaying of the prediction results.

Non Functional Requirements:

• The system is simple and easy to use.
• The system provides accurate results.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The general organization of the livestock behavior recognition system's hardware and software makeup, that is, its system architecture, defines how each component interacts with one another to use TinyML to classify livestock behavior. The three major components of this system are the dataset, the model training, the conversion of the model into TinyML-compatible form, and the prediction output.

First, a dataset of images of livestock under various conditions and behaviors are collected and used as input to train the machine learning model. Next, the trained model is used to identify the patterns of behavior in the dataset through the use of Python and TensorFlow Lite framework. Following training, the model is converted to a format compatible with TinyML for efficient operation. Finally, the system accepts new input data and predicts the behavior of the livestock based on the trained TinyML model; the output is then provided to the user. As such, the architecture allows for both efficient and effective livestock behavior identification using TinyML.

4.2 System Workflow

The system workflow defines the steps that describe the process of recognizing livestock behavior using tinyml.
Step 1: collecting dataset
A livestock image dataset is gathered and prepared for use in training a machine learning model. This dataset will contain images that represent a variety of livestock behaviors.

Step 2: prepare data
The data is gathered and prepared in a way to be able to train a machine learning model with it. The images are used to provide the model with input to learn with.
Step 3: train model
A machine learning model is trained using Python and the TensorFlow Lite framework. The model learns to identify patterns and features from the dataset it was trained on.
Step 4: convert to tinyml model format
Once trained the model is converted into tinyml format (TensorFlow Lite model). This will allow the model to be executed efficiently and lightly.
Step 5: predict behavior
The trained tinyml model will then take input data and make a prediction of what livestock behavior exists based on that input data.
Step 6: display results
The system displays the results of the predicted livestock behavior as an output to the user.

This workflow provides a reliable and accurate method of identifying livestock behaviors using tinyml.

4.3 UML Diagrams

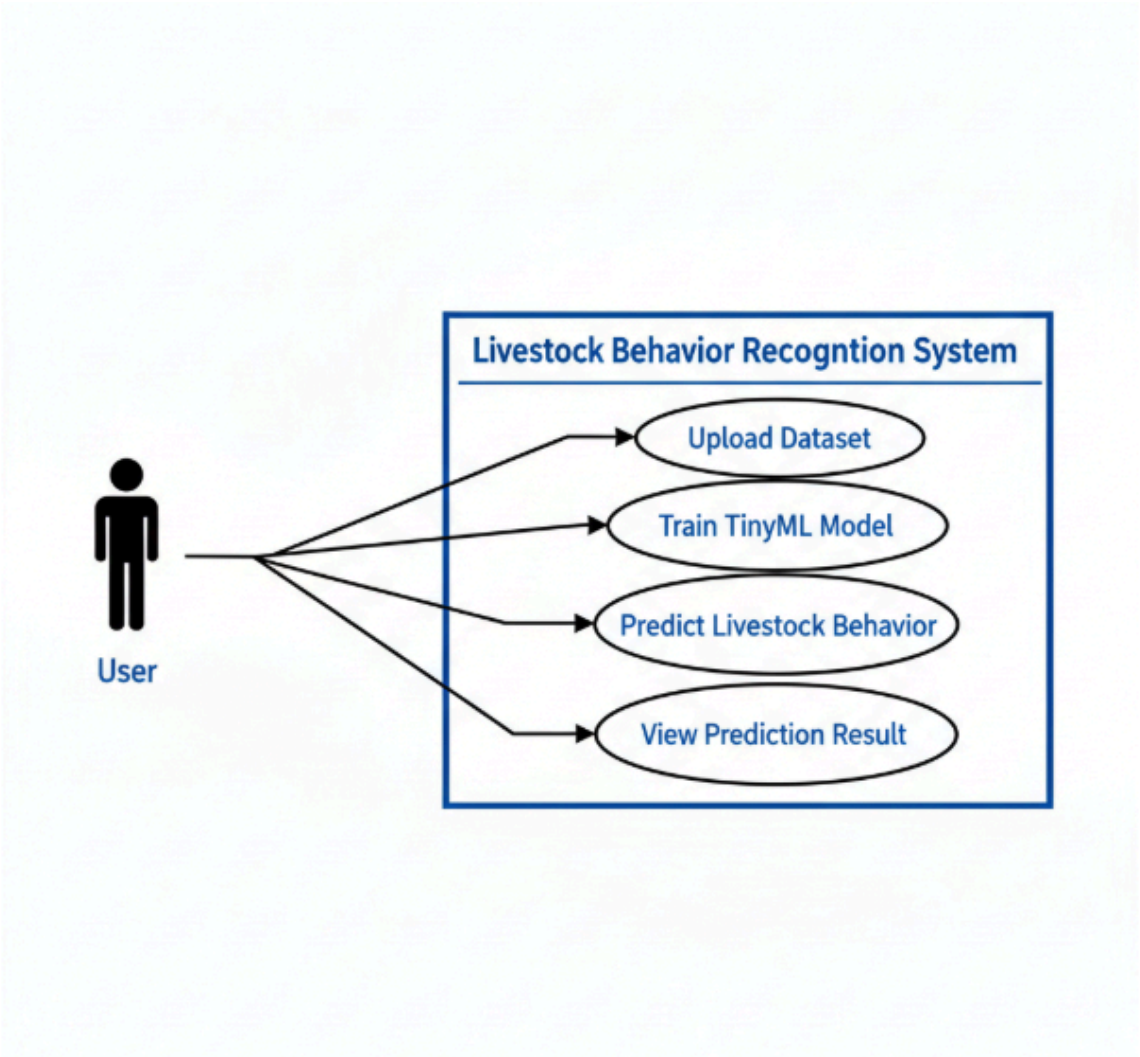

Figure 4.1 Use Case Diagram of Livestock Behavior Recognition System

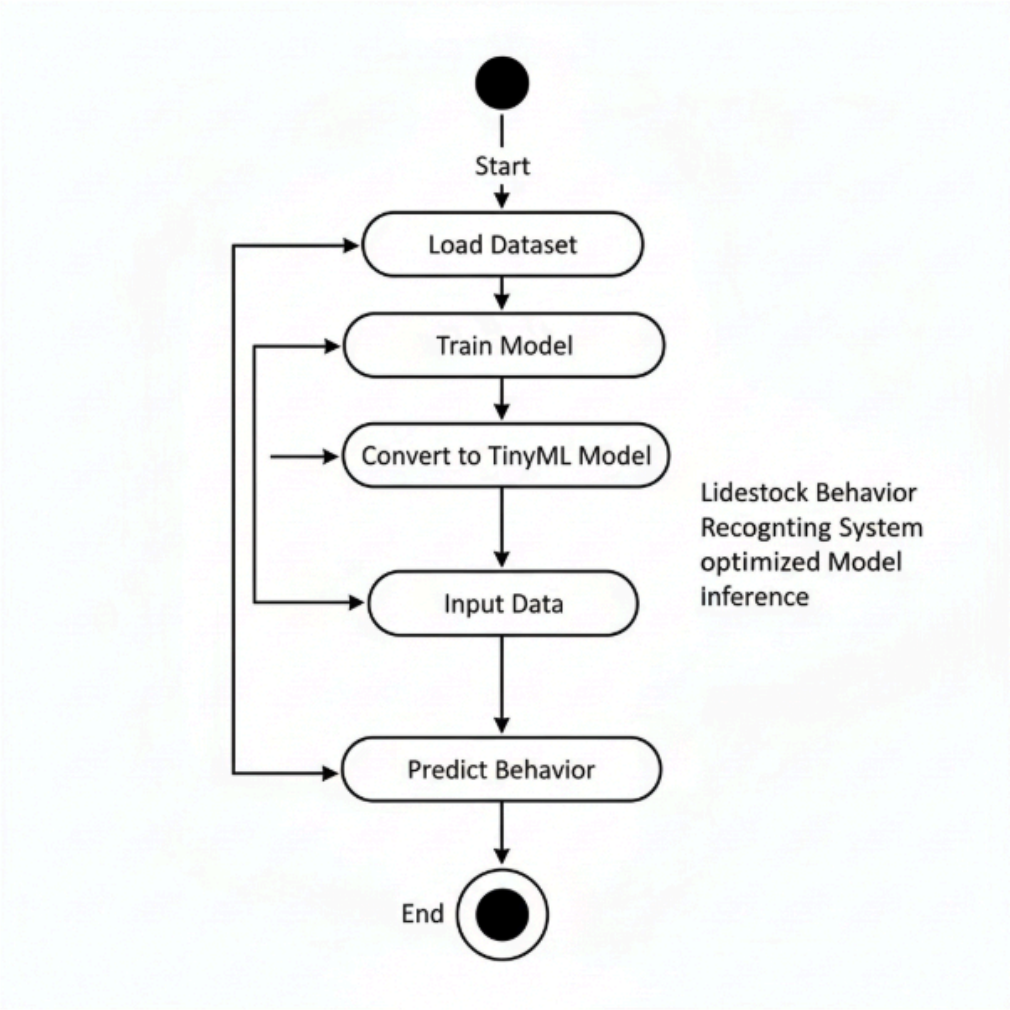

Figure 4.2 Activity Diagram of Livestock Behavior Recognition System

CHAPTER 5

IMPLEMENTATION

This chapter covers the methods and tools used to develop the Livestock Behavior Recognition System utilizing TinyML.

The Livestock Behavior Recognition System was developed with the use of Python programming language and TensorFlow Lite machine learning library, which were both utilized to train the livestock behavior recognition model as well as to evaluate it.

Python was utilized for the development of the software that trained the machine learning model. Python has many libraries available for machine learning and data processing that make it easy to write the code required for these tasks.

TensorFlow Lite was utilized to convert the machine learning model that had been trained into a TinyML model. The TinyML model is much lighter weight and more efficient than other models, this makes it more ideal for embedded and low resource computing devices.

Visual Studio Code and Jupyter Notebook were utilized as the development environments in which to write and run the Python code.

The Livestock Behavior Recognition System was developed and tested on a computer that ran the Windows operating system. The tools mentioned above created an efficient environment for developing and testing the Livestock Behavior Recognition System.

5.2 Module Description

The livestock behaviour recognition system has been broken down into many modules, with each module performing an individual task within the entire system.

Dataset Module:
This module is designed to load the livestock behaviour dataset. It will also be the module that manages the dataset. The dataset is made up of images used to both train and test the machine learning model.

Model Training Module:
This module is used to train the machine learning model. The model uses the patterns learned from the images of the livestock in order to predict the various behaviours of the livestock.

TinyML Conversion Module:
Once the machine learning model has been trained it can then be converted to TinyML through the use of TensorFlow Lite. Once the model is converted to TinyML, it becomes a much lighter weight and more efficient model.

Prediction Module:
This module will make predictions on livestock behaviour once the model has been trained to TinyML. The system inputs data and outputs the predicted behaviour.

Output Module:
This module will display the results of the prediction to the user. The results show what livestock behaviour was recognised by the system.

5.3 System Implementation

A livestock behavior recognition system was developed with the use of Python and the TensorFlow Lite framework.

The implementation steps included the process of:
1) Loading the livestock image dataset.
2) Training the model with the image dataset.
3) Converting the model into TinyML model format.
4) Behavior prediction by utilizing the TinyML model.

Firstly, the livestock image dataset was uploaded into the system. The livestock image dataset was used to develop a machine learning model that can recognize livestock behavior patterns.

After the model was developed, it was then converted into TinyML format, using TensorFlow Lite, which allows for a lightweight model that is very efficient in terms of processing power, which makes it ideal for behavior prediction.

Secondly, the TinyML model was utilized to predict livestock behavior from the input data received by the system. After processing the input, the system generates the predicted behavior as output.

Thirdly, the implementation was tested on a computer (using a Python-based development environment). The system was able to identify the behavior of the livestock and display the results of the predictions. The implementation of this system demonstrated that TinyML can be effective for recognizing livestock behavior.

5.4 Screenshots

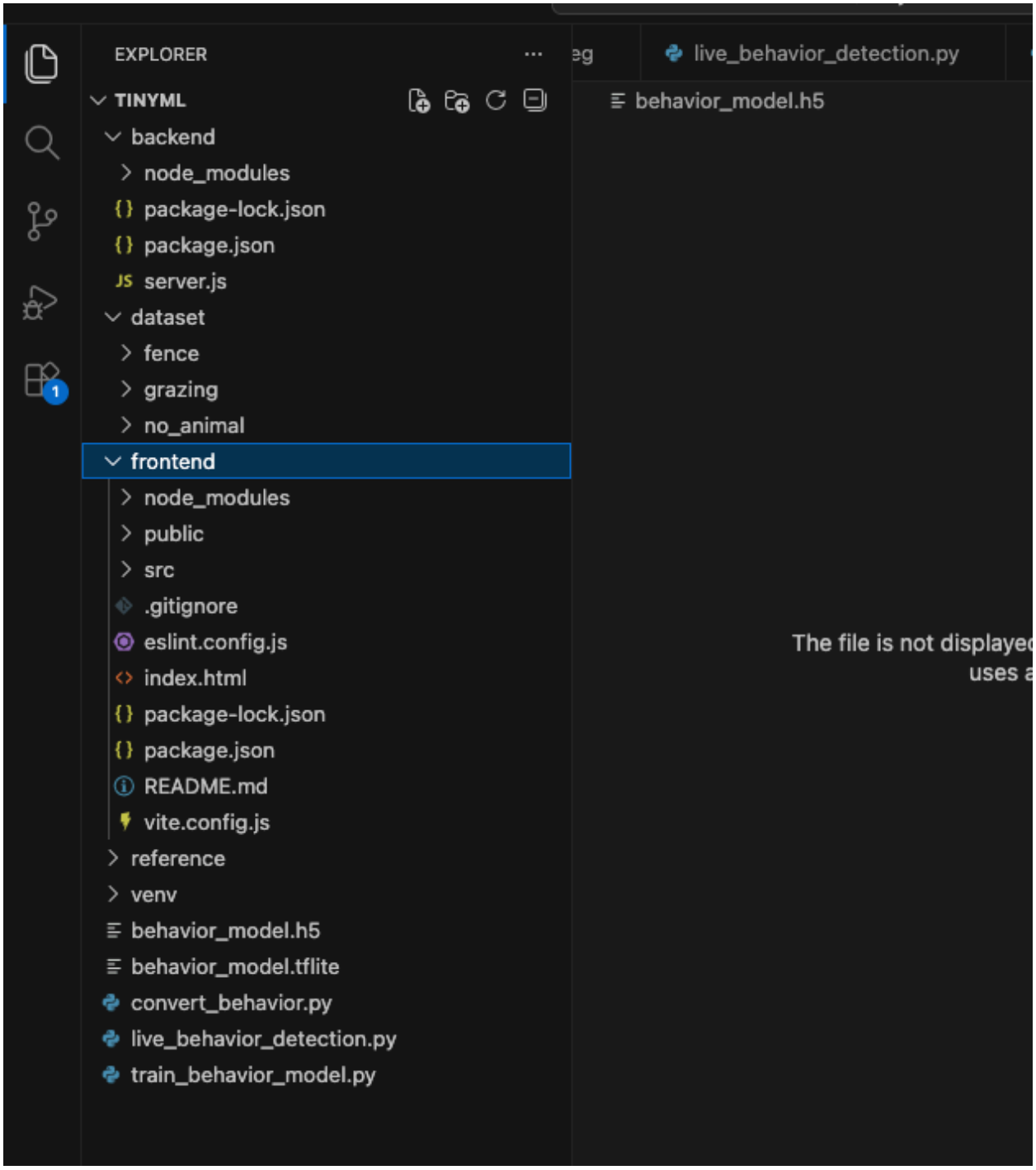

Figure 5.1 Dataset Folder Structure

Prediction Output Screenshot

Figure 5.2 Prediction Output – Fence Behavior Detected

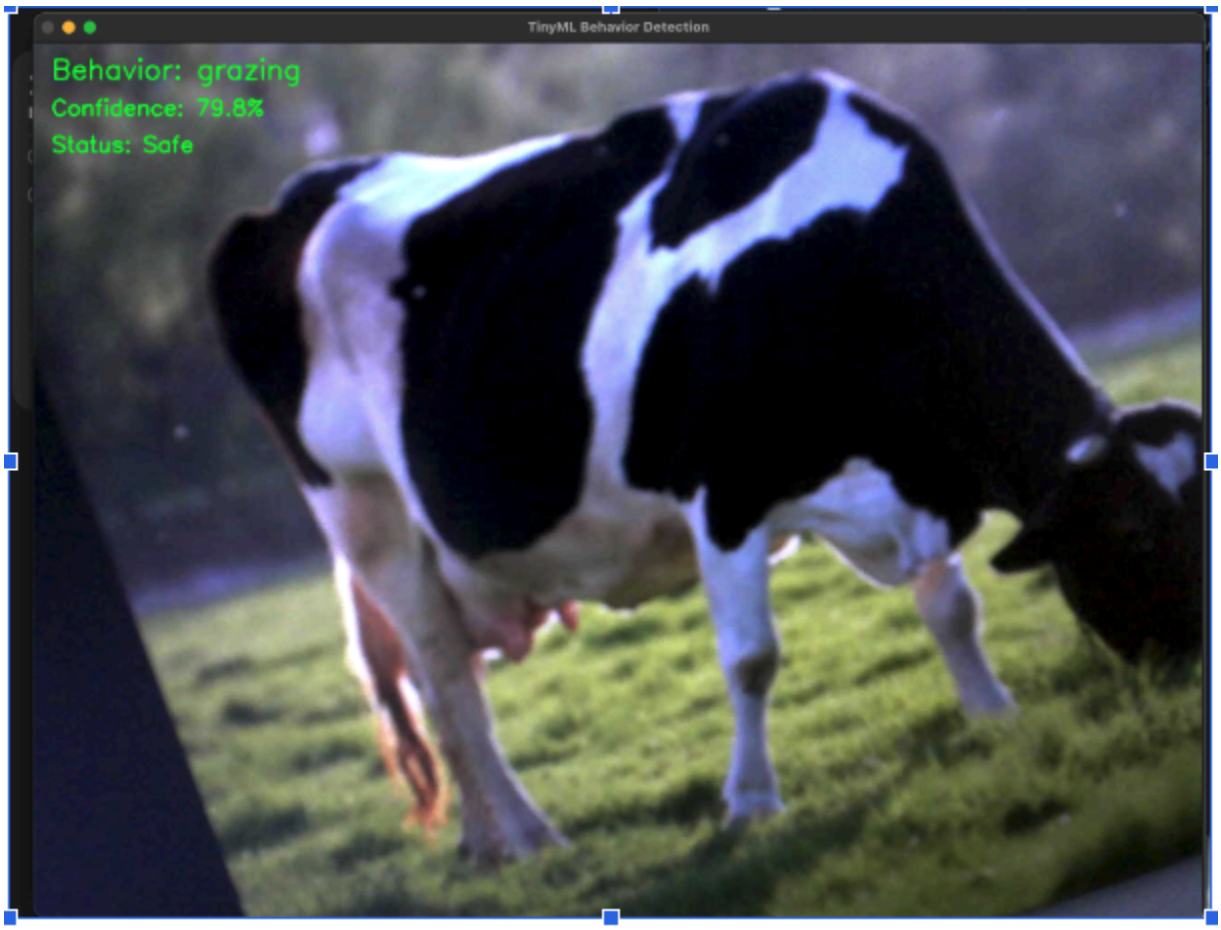

Figure 5.3 Prediction Output – Grazing Behavior Detected

Figure 5.4 Prediction Output – No Animal Detected

Behavior	Prediction Accuracy (%)
Fence	85%
Grazing	79%
No Animal	97%

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

A successful livestock behaviour recognition system was implemented and evaluated in Python using TensorFlow Lite. An image dataset of livestock was utilised to train a machine learning model; the model was then compiled as a TinyML version.

A TinyML model can determine the livestock behaviour from the provided input data; it has been shown that the model is capable of producing accurate predictions with reasonable efficiency. These predictions were made available to the end-user via the development environment.

The system has been subjected to testing using a variety of different input samples, and it has been demonstrated that the model was successful in classifying the livestock behaviour. This indicates that TinyML can be effectively applied to livestock behaviour recognition.

This illustrates that the system is able to provide a method for behaviour recognition using a lightweight machine learning model. Therefore, this system would be effective and suitable for use in resource-constrained environments.

6.2 Discussion

The livestock behavior recognition system was built with TinyML and Machine Learning. The TinyML enabled us to build an efficient model that could predict livestock behaviors based on the data set we were working from.

TinyML allowed us to create a lightweight and efficient machine learning model that produced fast and accurate predictions. It also showed that TinyML is a viable option when it comes to building efficient Behavior Recognition Systems.

We used Python and TensorFlow Lite as the base code for our implementation because they provided an effective platform to train and test our model. The prediction results show that our system can recognize the correct behavior of the livestock.

Our system overall worked very well and achieved all of the goals we had for this project. The results confirmed that TinyML can be utilized for the purposes of recognizing livestock behavior and can aid in the development of Intelligent Livestock Monitoring Systems.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The development of a livestock behavior recognition system by using TinyML and machine learning methods has been successful.

Python and TensorFlow Lite were used to implement the system.
A machine learning model was trained with images from the livestock image data set and then transformed into TinyML format.

TinyML was able to recognize livestock behavior, producing accurate prediction results.

This demonstrates that lightweight machine learning models can be applied effectively in animal behavior recognition.

Therefore this project met it's objectives of developing an efficient livestock behavior recognition system using TinyML.

This project is an example of how TinyML can be used to create intelligent and efficient animal behavior recognition systems.
The project presents a simple and effective approach to recognizing the behaviors of livestock.

7.2 Future Work

In addition to the enhancements already mentioned, there are several ways that the livestock behavior recognition system could continue to be improved into the future.

Additional data (larger and more diverse) could be collected and incorporated into the system to improve its accuracy.

The system could be built directly onto an embedded device such as a microcontroller to allow for the use of the system in a real time livestock monitoring setting.

More sensors and advanced machine learning models could be utilized to further increase the performance of the system.

Additionally, a graphical user interface could be created to provide a more user friendly way to interact with the system.

Finally, additional capabilities such as real-time monitoring and alert systems could be added to the system.

Together these enhancements would greatly improve the performance and usability of the livestock behavior recognition system.

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[4] Python Software Foundation, https://www.python.org